Pipe flow and transition to turbulence

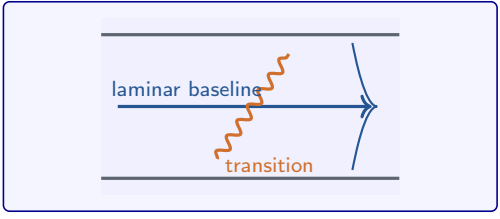
A controlled baseline for thinking about aortic flow

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IBFE Tutorials

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.



From a solvable baseline to aortic complexity

1. Derive Hagen–Poiseuille flow and audit every sign in Δp, Q, and wall shear.
2. Nondimensionalize with Re_D and distinguish laminar drag from turbulent statistics.
3. Reconcile modal stability with transient growth and finite-amplitude transition.
4. Read puffs, slugs, decay, and splitting in space–time.
5. Add pulsatility, curvature, branching, compliance, and inflow disturbances.

The straight rigid pipe is a reference problem. Its transition thresholds do *not* transfer directly to the aorta.

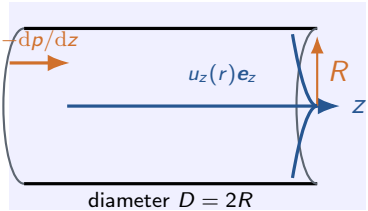
Fix the geometry and Reynolds-number convention

Use cylindrical coordinates (z, r, θ) with

$$D = 2R, \qquad \nu = \frac{\mu}{\rho},$$

$$\mathrm{Re}_D := \frac{U_b D}{\nu} = \frac{\rho U_b D}{\mu}.$$

Here $U_b = Q/(\pi R^2)$ is the bulk velocity.



Why the subscript matters

Pipe-transition references also use radius, centerline velocity, or friction velocity. State the convention before quoting a Reynolds number.

The laminar baseline is deliberately restrictive

Straight-pipe model

- incompressible, Newtonian, constant ρ, μ;
- rigid, circular, straight wall;
- no slip;
- steady, axisymmetric, fully developed flow.

Aortic departures

- pulsatile forcing and finite entrance length;
- curvature, branches, taper, noncircular sections;
- compliant, prestressed, anisotropic wall;
- valve jet, swirl, and upstream disturbances.

Strategy: derive the reference solution exactly, then identify which omitted mechanism changes each conclusion.

Start from dimensional incompressible Navier–Stokes

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mu\nabla^2\boldsymbol{u},$$
$$\nabla\cdot\boldsymbol{u} = 0.$$

Inertial terms

$\rho\partial_t\boldsymbol{u}$ is local acceleration; $\rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ is nonlinear convection.

Forcing and diffusion

$-\nabla p$ drives the flow; $\mu\nabla^2\boldsymbol{u}$ diffuses momentum.

Density is constant here, and no body force is included. Gravity can be absorbed into a modified pressure when appropriate.

Fully developed symmetry removes the nonlinear term

Take

$$\boldsymbol{u} = u_z(r)\boldsymbol{e}_z, \qquad \partial_t u_z = \partial_z u_z = \partial_\theta u_z = 0.$$

Continuity is automatic, and $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = u_z\partial_z(u_z\boldsymbol{e}_z) = \boldsymbol{0}$.
The axial momentum equation becomes

$$0 = -\frac{\mathrm{d}p}{\mathrm{d}z} + \mu\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u_z}{\mathrm{d}r}\right).$$

For a constant pressure drop, $\mathrm{d}p/\mathrm{d}z < 0$. Define the positive driving gradient $g := -\mathrm{d}p/\mathrm{d}z > 0$.

Integrate the radial ODE with two physical conditions

With $g = -\mathrm{d}p/\mathrm{d}z$,

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u_z}{\mathrm{d}r}\right) = -\frac{g}{\mu}.$$

Integrating once gives

$$r\frac{\mathrm{d}u_z}{\mathrm{d}r} = -\frac{g}{2\mu}r^2 + C_1.$$

Regularity at the centerline

$u_z'(0) = 0$ and boundedness force $C_1 = 0$.

No slip at the wall

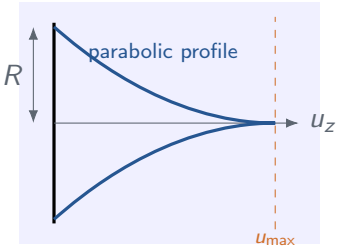
$u_z(R) = 0$ determines the remaining constant.

Hagen–Poiseuille flow is parabolic

$$\boxed{u_z(r) = \frac{g}{4\mu}(R^2 - r^2)}.$$

Checks:

- $u_z(R) = 0$ and $u_z'(0) = 0$;
- $u_z(r) \geq 0$ because $g > 0$;
- $u_z'' < 0$: the centerline is the maximum;
- units: $[gR^2/\mu] = L/T$.



Integrating the profile gives Q, U_b, and $u_{\max}$

$$Q = \int_0^R 2\pi r\, u_z(r)\, \mathrm{d}r = \frac{\pi g R^4}{8\mu},$$

$$U_b := \frac{Q}{\pi R^2} = \frac{gR^2}{8\mu},$$

$$u_{\max} = u_z(0) = \frac{gR^2}{4\mu} = 2U_b.$$

Therefore

$$u_z(r) = 2U_b \left(1 - \frac{r^2}{R^2}\right).$$

The bulk velocity U_b, not the centerline velocity, defines Re_D in this deck.

Separate signed wall traction from its positive magnitude

For a Newtonian fluid, $\boldsymbol{\sigma} = -p\boldsymbol{I} + \mu(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T)$. At the cylindrical wall, the outward normal from the fluid is $\boldsymbol{n} = \boldsymbol{e}_r$:

$$(\boldsymbol{\sigma}\boldsymbol{n}) \cdot \boldsymbol{e}_z = \mu u_z'(R) = -\frac{gR}{2} = \frac{R}{2}\frac{\mathrm{d}p}{\mathrm{d}z} < 0.$$

This signed traction opposes the flow.
Define the **positive wall-shear magnitude**

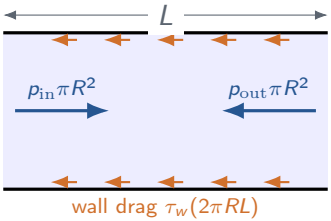
$$\boxed{\tau_w := -\mu u_z'(R) = \frac{gR}{2} = -\frac{R}{2}\frac{\mathrm{d}p}{\mathrm{d}z} > 0.}$$

Use this τ_w in friction-factor formulas.

A control-volume balance confirms the sign

For a pipe segment of length L,

$$\underbrace{\Delta p\,\pi R^2}_{\text{pressure force downstream}} = \underbrace{\tau_w (2\pi R L)}_{\text{wall drag upstream}} .$$



Hence

$$\tau_w = \frac{\Delta p\, R}{2L} = \frac{\Delta p\, D}{4L}, \qquad \Delta p = gL.$$

Combining with $U_b = gR^2/(8\mu)$ gives

$$\Delta p = \frac{8\mu L U_b}{R^2} = \frac{32\mu L U_b}{D^2}.$$

Darcy and Fanning friction factors differ by four

The Darcy–Weisbach convention is

$$\Delta p = f_D \frac{L}{D} \frac{\rho U_b^2}{2}, \qquad f_D = \frac{8\tau_w}{\rho U_b^2}.$$

For Hagen–Poiseuille flow,

$$\boxed{f_D = \frac{64}{\mathrm{Re}_D}}.$$

The Fanning convention instead defines

$$f_F = \frac{\tau_w}{\frac{1}{2}\rho U_b^2} = \frac{16}{\mathrm{Re}_D}, \qquad f_D = 4 f_F.$$

Notation check

Always identify the convention: the same symbol f is used for both in the literature.

Nondimensionalization isolates Re_D

Choose

$$\boldsymbol{x} = D\boldsymbol{x}^*, \qquad t = \frac{D}{U_b}t^*, \qquad \boldsymbol{u} = U_b\boldsymbol{u}^*, \qquad p = \rho U_b^2 p^*.$$

After substitution and dropping stars,

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}_D}\nabla^2\boldsymbol{u}, \qquad \nabla\cdot\boldsymbol{u} = 0.$$

$\mathrm{Re}_D = U_b D/\nu$ compares convective inertia with viscous diffusion under this scaling. It is essential, but it is not the only control parameter once forcing, geometry, and wall mechanics vary.

Turbulence changes drag and introduces fluctuations

At *fixed bulk flow rate*, turbulence generally requires a larger pressure gradient than the laminar relation predicts; the friction factor departs from $64/\mathrm{Re}_D$.

For smooth turbulent pipes the Blasius correlation $f_D \approx 0.316\,\mathrm{Re}_D^{-1/4}$ ($4\times10^3 \lesssim \mathrm{Re}_D \lesssim 10^5$) applies; rough pipes follow the Colebrook equation, summarized in the Moody chart.

At fixed pressure gradient, the bulk flow rate instead changes. State what is held fixed.

Choose a time, ensemble, or phase average and write

$$\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}', \qquad \overline{\boldsymbol{u}'} = 0.$$

Although the mean fluctuation vanishes,

$$\overline{u_i' u_j'} \neq 0$$

and these correlations alter mean momentum transport.

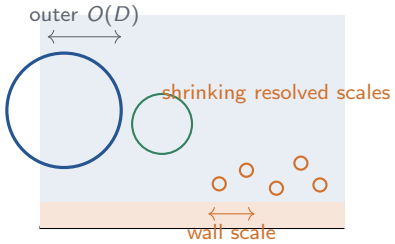
For pulsatile flow, a phase average and a long-time mean answer different questions.

DNS cost grows after turbulence creates small scales

Reynolds number alone does not create a turbulent cascade: the flow must first become turbulent.
Once it does,

- outer structures remain $O(D)$;
- dissipative and near-wall scales shrink;
- the wall-normal mesh and time step become severe;
- wall-resolved cost is often organized by the friction Reynolds number $\mathrm{Re}_\tau = u_\tau R/\nu$, with friction velocity $u_\tau = \sqrt{\tau_w/\rho}$.

Exact cost scaling depends on domain, numerics, and what is resolved.



Pipe transition has no known modal instability

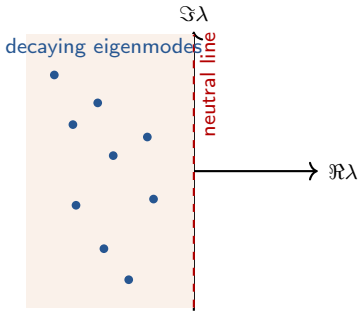
Fully developed Hagen–Poiseuille flow is eigenvalue stable in all computations to date. Meseguer and Trefethen computed a stable linearized spectrum through

$$\mathrm{Re}_D = 10^7.$$

Yet finite disturbances can trigger turbulence at far lower Reynolds numbers.

Subcritical transition

Infinitesimal eigenmodes decay, but finite-amplitude perturbations, transient amplification, and nonlinear feedback can produce turbulent dynamics.



Meseguer & Trefethen (2003)

Perturb about the Poiseuille base state

Let $\boldsymbol{u} = U(r)\boldsymbol{e}_z + \widetilde{\boldsymbol{u}}$. The nondimensional perturbation equation is

$$\partial_t \widetilde{\boldsymbol{u}} + U\partial_z \widetilde{\boldsymbol{u}} + \widetilde{u}_r U'(r)\boldsymbol{e}_z + (\widetilde{\boldsymbol{u}} \cdot \nabla)\widetilde{\boldsymbol{u}} = -\nabla \widetilde{p} + \frac{1}{\mathrm{Re}_D}\nabla^2 \widetilde{\boldsymbol{u}}, \qquad \nabla \cdot \widetilde{\boldsymbol{u}} = 0.$$

- Linear dynamics include base advection, shear coupling, pressure, and viscosity.
- The quadratic term redistributes energy among perturbation modes.
- Use $\widetilde{\boldsymbol{u}} = 0$ at $r = R$ and periodic or suitably decaying conditions in z.

Three different tildes and primes

Here $\widetilde{\boldsymbol{u}}$ is a *deterministic* perturbation about a specified base state, not the turbulent fluctuation $\boldsymbol{u}'$ of the Reynolds decomposition (a statistical deviation from an average, as on the turbulence slide), and not the LES filtered field $\widetilde{\boldsymbol{u}}$ of deck 5-3 (a spatially coarse-grained field). Same symbols, three different meanings.

Non-normality permits transient growth

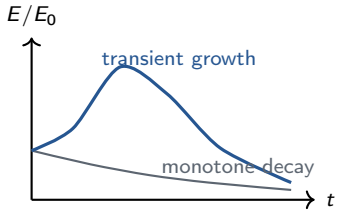
For the linearized system $\partial_t \boldsymbol{q} = L\boldsymbol{q}$,

$$L^*L \neq LL^*, \qquad G(t) := \max_{\boldsymbol{q}_0 \neq 0} \frac{\|e^{tL}\boldsymbol{q}_0\|_E^2}{\|\boldsymbol{q}_0\|_E^2}.$$

Stable eigenvalues do not require $G(t)$ to decrease monotonically.

High-Re_D scaling

For the optimal streamwise-constant, azimuthal-$n = 1$ mechanism, velocity amplification is $O(\mathrm{Re}_D)$ and energy gain is $O(\mathrm{Re}_D^2)$.

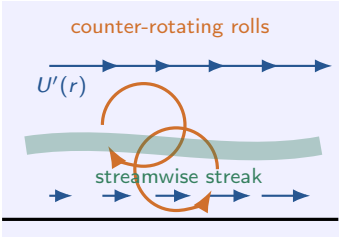


Transient growth is an amplification mechanism, not by itself a transition criterion.

Lift-up converts cross-stream motion into streaks

Weak streamwise vortices move low-speed fluid away from the wall and high-speed fluid toward it. The base shear then creates large streamwise velocity streaks.

- cross-stream velocity can remain small;
- the induced streamwise perturbation can grow algebraically;
- nonlinear feedback is needed to regenerate the rolls and sustain turbulence.



The perturbation-energy budget isolates mean-shear extraction

Let

$$E(t) = \frac{1}{2} \int_\Omega |\widetilde{\boldsymbol{u}}|^2 \, \mathrm{d}V.$$

With no slip at the wall and periodic or suitably decaying axial conditions,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\int_\Omega \widetilde{u}_z \widetilde{u}_r U'(r) \, \mathrm{d}V - \frac{1}{\mathrm{Re}_D} \int_\Omega |\nabla \widetilde{\boldsymbol{u}}|^2 \, \mathrm{d}V.$$

The first term is signed production from the base shear; the second is nonnegative viscous dissipation.

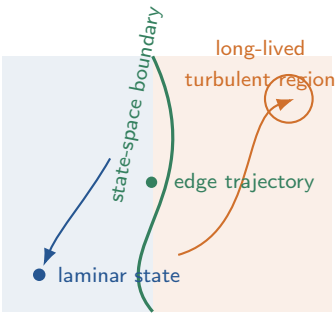
The nonlinear perturbation term redistributes energy but contributes zero to the net integral under these conditions.

Above an energy-stability bound, the estimate becomes inconclusive; it does not prove transition.

Finite-amplitude outcomes motivate an “edge” picture

At a fixed Reynolds number and observation protocol:

- sufficiently small disturbances return to laminar flow;
- some larger disturbances visit long-lived turbulent dynamics;
- a complicated state-space boundary separates these outcomes.



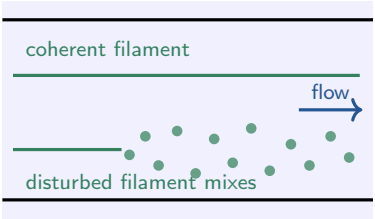
Scope of the picture

In a finite or open pipe, “sustained” depends on domain length, forcing, and observation time. The edge is a dynamical-systems idealization, not a universal amplitude threshold.

Reynolds' dye experiment also measures disturbance sensitivity

At low Re_D, a carefully introduced filament can remain coherent. With larger Reynolds number or larger inlet disturbance, it breaks down and mixes.

- transition is not specified by Re_D alone;
- inlet quality, disturbance shape, pipe length, and observation time matter;
- deliberately triggered and naturally occurring transition should be distinguished.



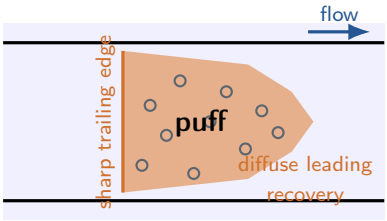
Reynolds (1883)

A puff is localized—not immortal

A puff is a localized turbulent patch in the lower transitional range.

- sharp upstream/trailing interface;
- diffuse downstream/leading recovery region;
- travels near the bulk velocity;
- remains approximately localized while it exists.

An individual puff can decay stochastically; "persistent" should not mean infinite lifetime.



A slug grows because its two fronts have different speeds

A turbulent slug has identifiable leading and trailing fronts. In the laboratory frame, both fronts normally advect downstream, but

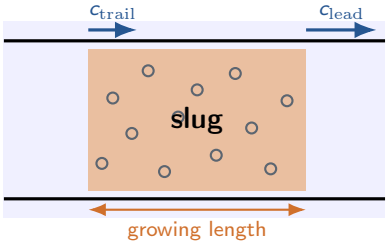
$$c_{\text{lead}} > c_{\text{trail}}$$

so the turbulent region lengthens.

Wording to avoid

"The slug propagates upstream" confuses growth relative to the fluid with absolute front motion.

Puffs and slugs describe different transitional regimes, not merely different drawing styles.



Anatomy and front behavior: Wygnanski & Champagne (1973).

Puff decay and splitting are competing stochastic events

Measure two conditional times at fixed Re_D and disturbance protocol:

$$\tau_{\mathrm{decay}}(\mathrm{Re}_D), \qquad \tau_{\mathrm{split}}(\mathrm{Re}_D).$$

Decay dominates

If splitting is slower than decay, turbulent patches disappear in the long-pipe, long-time limit.

Splitting dominates

If splitting is faster, new puffs can maintain a nonzero turbulent fraction even though each puff remains mortal.

Avila et al. measured a balance at

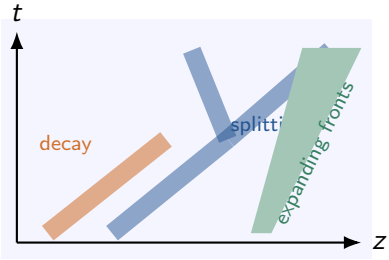
$$\boxed{\mathrm{Re}_D = 2040 \pm 10.}$$

This is a statistical onset of sustained turbulence, not an immortality threshold for one puff. The textbook value $\mathrm{Re}_D \approx 2300$ marks practical onset of transition; 2040 is the statistically defined point where puff splitting balances decay.

A space–time diagram reveals propagation, decay, and splitting

Horizontal position is z; time increases upward. The slope of each interface encodes a front speed.

- a band ending in time is puff decay;
- one band branching into two is splitting;
- widening separation between fronts is slug growth;
- turbulent fraction becomes a spatiotemporal statistic.



Critical balance: Avila et al. (2011).

Transition observables need a conditioning statement

Useful measurements include

- puff decay and splitting time distributions;
- leading/trailing front speeds and turbulent fraction;
- probability of relaminarization over a stated observation window;
- friction-factor departure from $64/\mathrm{Re}_D$;
- velocity variance, Reynolds stresses, spectra, and wall-shear statistics.

For smooth turbulent pipes the Blasius correlation $f_D \approx 0.316\,\mathrm{Re}_D^{-1/4}$ ($4 \times 10^3 \lesssim \mathrm{Re}_D \lesssim 10^5$) applies; rough pipes follow the Colebrook equation, summarized in the Moody chart.

Report Re_D, pipe length, inlet/disturbance protocol, boundary conditions, and observation time. One finite run cannot establish "sustained forever."

For cardiovascular flow, connect each statistic to pressure loss, wall loading, mixing, or another biomechanical quantity of interest.

The pipe-to-aorta analogy names mechanisms, not thresholds

What transfers

- inertia–viscosity competition;
- shear production and transient growth;
- disturbance sensitivity;
- intermittency and front dynamics.

What changes

- periodic acceleration/deceleration;
- curvature, branches, entrance jets, taper;
- moving compliant walls and pressure waves;
- finite length and patient-specific inflow.

Consequence: a steady straight-pipe transition Reynolds number cannot be imported as an aortic turbulence criterion.

Womersley number compares unsteadiness with viscous diffusion

For angular frequency ω,

$$\alpha = R\sqrt{\frac{\omega}{\nu}} = R\sqrt{\frac{\omega\rho}{\mu}}.$$

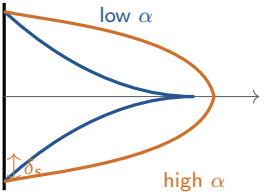
The oscillatory Stokes-layer thickness is

$$\delta_s = \sqrt{\frac{2\nu}{\omega}}, \qquad \frac{\delta_s}{R} = \frac{\sqrt{2}}{\alpha}.$$

If $\mathrm{St}_\omega = \omega D/U_b$, then

$$\alpha^2 = \frac{\mathrm{Re}_D \mathrm{St}_\omega}{4}.$$

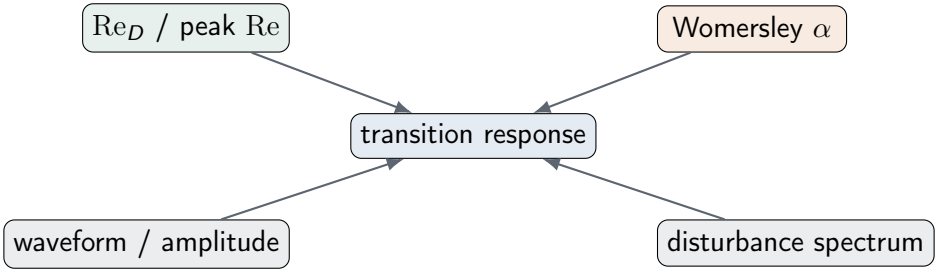
Large α means a thinner oscillatory wall layer and stronger phase lag; it does not imply turbulence.



profiles shown schematically at one phase

Womersley (1955)

Pulsatile transition occupies a multi-parameter space



Acceleration, deceleration, and flow reversal can change the base profile and its receptivity throughout a cycle. A phase-resolved stability or simulation analysis must specify the waveform and disturbance.

Neither high peak Reynolds number nor large Womersley number alone is a turbulence criterion.

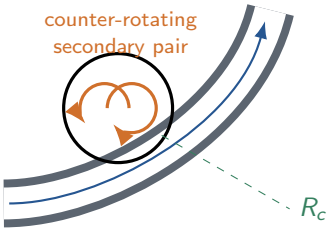
Curvature introduces secondary flow; branching is different

With centerline radius of curvature R_c, define

$$\delta_c = \frac{R}{R_c} = \frac{D}{2R_c}, \qquad \mathrm{De} = \mathrm{Re}_D \sqrt{\delta_c}.$$

The curvature ratio is written δ_c, not κ, because κ is reserved for the von Kármán constant. This diameter-based convention for δ_c must be stated because alternatives exist.

In a simple steady laminar curved pipe, centrifugal effects can drive a counter-rotating Dean-vortex pair. Finite curvature, torsion, pulsatility, and wall motion add parameters.



Dean (1927)

Branching caveat

A branch-induced jet or separation zone is not a Dean vortex.

Compliance adds wave propagation; inflow sets disturbance amplitude

For normalized area distensibility

$$\mathcal{D}_A = \frac{1}{A}\frac{\mathrm{d}A}{\mathrm{d}p}, \qquad c = \frac{1}{\sqrt{\rho \mathcal{D}_A}}.$$

This is the **Bramwell–Hill** relation (Bramwell & Hill 1922, *Proc. R. Soc. B* **93**, 298). A thin, linearly elastic, untethered tube gives the **Moens–Korteweg** scale (Moens 1878; Korteweg 1878)

$$c_0 \approx \sqrt{\frac{Eh}{\rho D}}, \qquad \frac{U_b}{c_0}$$

for convective speed relative to wave speed.

Real arteries are prestressed, tethered, anisotropic, viscoelastic, and nonlinear. The specified FSI model determines whether compliance amplifies or attenuates a disturbance.

Valve jets, swirl, upstream turbulence, and phase-dependent inlet profiles set finite-amplitude perturbations that a subcritical system can amplify.

State the wall law and inlet condition before making a stability claim.

Choose a model by resolved physics and quantity of interest

Method	Resolved field	Key modeling	Aortic question
DNS	all dynamically relevant scales	geometry, wall, inflow	reference cases; mechanism discovery
LES	large unsteady eddies	SGS, wall, inflow, FSI	jets, wakes, mixing, fluctuation statistics
RANS/URANS	mean or phase-resolved mean	Reynolds stress, wall, transition	design sweeps; mean pressure and shear
Hybrid	RANS region plus resolved eddies	grid/model interface, wall	high-Re separated regions

There is no universal accuracy ordering. Grid resolution, wall treatment, inflow data, FSI assumptions, and validation against the chosen output all matter.

Takeaways: carry conventions and caveats forward

- Hagen–Poiseuille flow links Δp, Q, U_b, and the *positive* wall-shear magnitude τ_w; Darcy and Fanning factors differ by four.
- Fully developed pipe flow has no known modal instability, yet non-normal growth can amplify finite disturbances before nonlinear feedback acts.
- Puffs can decay or split; sustained turbulence is a statistical population balance, not an immortal-puff threshold.
- Large Re_D or α alone does not establish turbulence in a pulsatile flow.
- Curvature, branching, compliance, and inlet disturbances require explicit additional parameters and models.

The pipe is the clean control problem; the aorta is a time-dependent fluid–structure system.

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